

AURA AI ASSISTANT — FULL ROADMAP (STEP 0 → PRODUCTION)

NOW → UPDATED ROADMAP (AURA AI V5 → V6 TRANSFORMATION)

(Based on your file — upgraded, corrected, and made production-ready)

STEP 0 — STRATEGIC FOUNDATION (DONE)

You already completed:

- AURA V5 dataset 
 - Fusion pipeline 
 - 1.25M rows 
 - LLM research 
-

SYSTEM EVOLUTION

AURA V5 → Data Intelligence System
AURA V6 → AI Decision Engine

UPDATED GLOBAL ARCHITECTURE

```
User / Admin
↓
FastAPI Backend
↓
AURA AI Orchestrator
↓
Qwen 7B (LOCAL BRAIN)
↓
Tool Layer (SQL / Analytics)
↓
AURA V5 Dataset (1.25M rows)
```

PHASE 1 — LLM DEPLOYMENT (LOCAL BRAIN)

Goal

Run Qwen locally and make it stable

Tasks

- Install runtime (Ollama or Transformers)
- Load Qwen 7B (quantized)
- Test:

```
→ reasoning
→ SQL generation
→ structured outputs
```

Output

```
LLM running locally and responding correctly
```

PHASE 2 — TOOL LAYER (MOST IMPORTANT)

Goal

Connect LLM to AURA dataset

Build:

SQL Engine

```
defquery_aura(sql):  
    returndb.execute(sql)
```

Analytics Engine

```
defanalyze_market(data):  
    returntrend,volatility,signals
```

Feature Access

```
defget_grid_data(grid_id,year,quarter):  
    returndf_filtered
```

Core Flow

```
LLM → SQL → dataset → result → explanation
```

 THIS is your real intelligence

PHASE 3 — LLM ORCHESTRATION (CORE ENGINE)

Goal

Turn Qwen into AURA AI

Add:

- system prompt (AURA rules)
 - structured outputs
 - tool-calling routing
-

Flow

```
User → LLM
LLM → decide action
→ SQL / analysis
→ result
→ explanation
```

Example

```
User:
"Best investment area in Lisbon?"

LLM:
→ query Cat7 + Cat4 + Cat6
→ compute ROI
→ explain reasoning
```

PHASE 4 — DOMAIN ADAPTATION (AURA BRAIN)

Goal

Make Qwen think like AURA

DO THIS (lightweight):

✓ Prompt engineering (MAIN)

Teach:

- real estate logic
 - AURA signals
 - decision rules
-

✓ Optional LoRA (only if time allows)

👉 Not required for success

PHASE 5 — EXPLAINABILITY SYSTEM

Goal

Make outputs trustworthy

Add:

- SHAP (admin insights)
 - LLM explanation layer
-

Output

"Price increased due to supply pressure + demographic shift"

PHASE 6 — (OPTIONAL) DUAL MODEL SYSTEM

 Only if time allows

Architecture

Gemma → chat layer
Qwen → reasoning engine

 You can skip this for internship

PHASE 7 — DEPLOYMENT (MINIMAL VIABLE SYSTEM)

Goal

Make system usable

Stack

- FastAPI (backend)
- Qwen (local)
- MySQL / CSV (dataset)
- simple UI OR API endpoints

Flow

Frontend → API → LLM → Tools → Dataset → Response

PHASE 8 — OPTIMIZATION (ONLY IF TIME)

Improve:

- latency
- prompt efficiency
- caching

PHASE 9 — FUTURE (AURA V6 / ENTERPRISE)

Upgrade path

Qwen 7B → Qwen MoE (37B cloud)

Future:

- cloud deployment
- real-time data
- multi-agent system

UPDATED TIMELINE (REALISTIC)

WEEK PLAN

Week 1

- Run Qwen locally
-

Week 2

- Build SQL tool
-

Week 3

- Connect LLM + dataset
-

Week 4

- Add reasoning + explanations
-

Week 5

- Build API / demo
-

Week 6

- polish + presentation
-



FINAL SYSTEM (WHAT YOU WILL HAVE)

```
User asks question
↓
LLM understands intent
↓
LLM generates SQL
↓
Dataset returns real data
↓
LLM explains insight
```